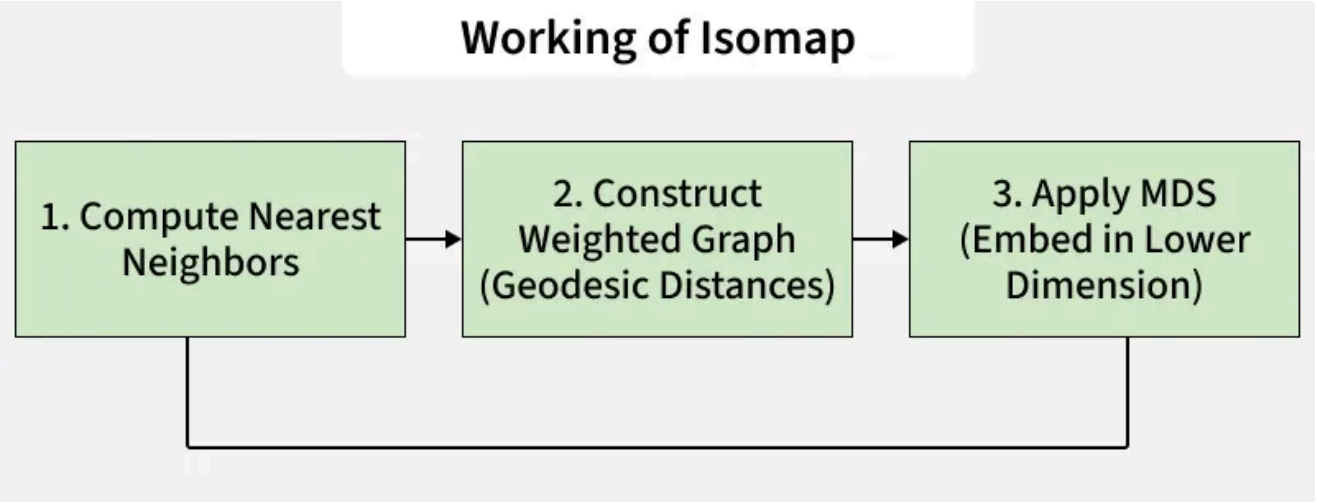
Isomap

It create a graph by connecting each instance to **its nearest neighbors**, then reduce dimensionality while trying to preserve the **gradient distances between the instance.**

Isomap (Isometric Mapping) is a **non-linear dimensionality reduction technique** that maps high-dimensional data into a lower-dimensional space while **preserving the "geodesic" (intrinsic, manifold-based)** distance between points.

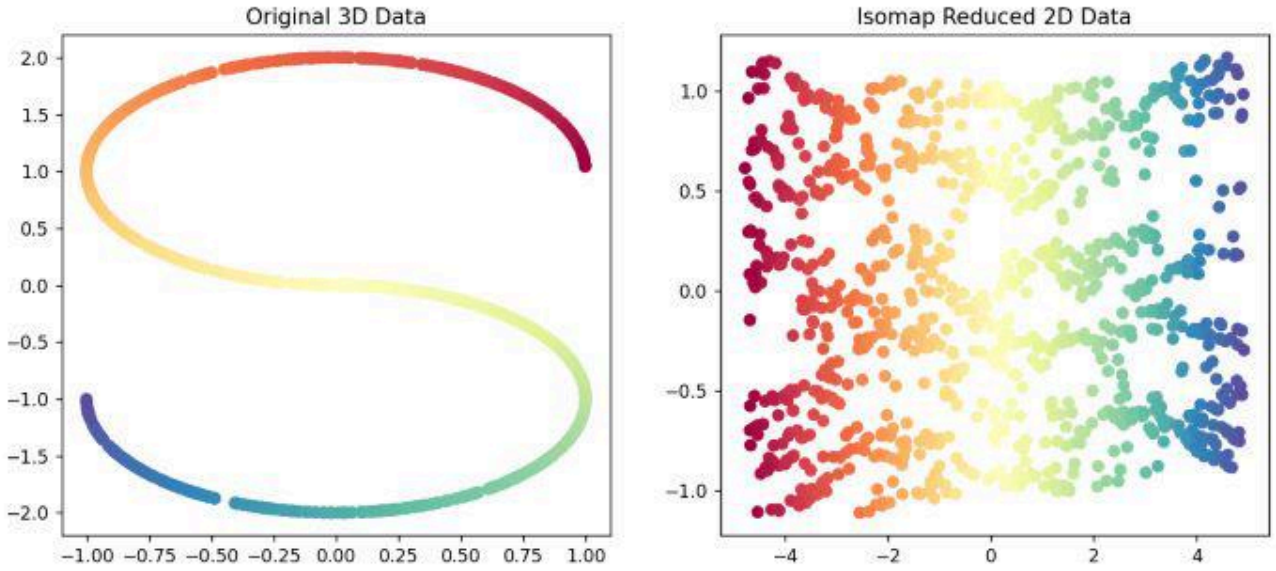
- Handles non-linear datasets better than PCA
- Preserves true manifold-based distances
- Helps visualize high-dimensional patterns
- Useful for images, shapes and other curved data



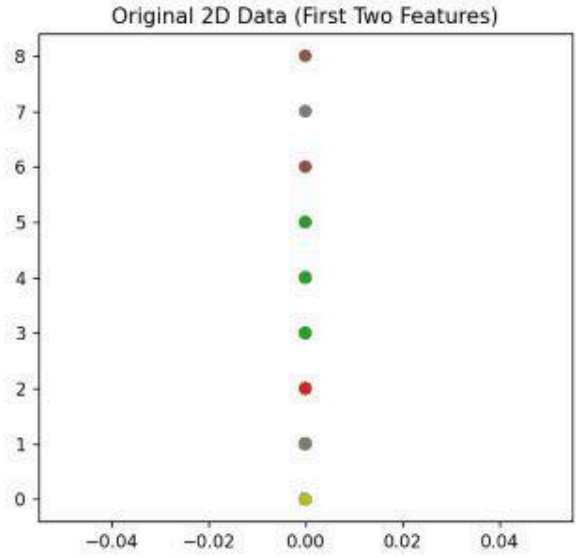
- **Geodesic Distance** is the shortest path between two points along the manifold's surface
- **Euclidean Distance*** is the straight-line distance between two points in the original space.

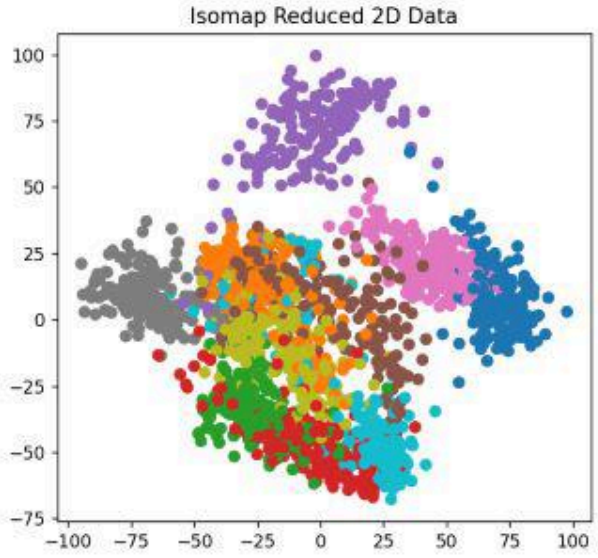
```
from sklearn.datasets import make_s_curve
from sklearn.manifold import Isomap
import matplotlib.pyplot as plt

X, color = make_s_curve(n_samples=1000, random_state=42)

isomap = Isomap(n_neighbors=10, n_components=2)
X_isomap = isomap.fit_transform(X)

fig, ax = plt.subplots(1, 2, figsize=(12, 5))

ax[0].scatter(X[:, 0], X[:, 2], c=color, cmap=plt.cm.Spectral)
ax[0].set_title('Original 3D Data')

ax[1].scatter(X_isomap[:, 0], X_isomap[:, 1], c=color, cmap=plt.cm.Spectral)
```

```
ax[1].set_title('Isomap Reduced 2D Data')

plt.show()
```




```
from sklearn.datasets import load_digits
from sklearn.manifold import Isomap
import matplotlib.pyplot as plt

digits = load_digits()

isomap = Isomap(n_neighbors=30, n_components=2)
digits_isomap = isomap.fit_transform(digits.data)

fig, ax = plt.subplots(1, 2, figsize=(12, 5))

ax[0].scatter(digits.data[:, 0], digits.data[:, 1], c=digits.target,
cmap=plt.cm.tab10)
ax[0].set_title('Original 2D Data (First Two Features)')

ax[1].scatter(digits_isomap[:, 0], digits_isomap[:, 1], c=digits.target,
cmap=plt.cm.tab10)
ax[1].set_title('Isomap Reduced 2D Data')

plt.show()
```





Pros:

- Visualisation
- Data Exploration
- Anomaly Detection
- Pre-processing for machine learning
- Cap tures non-linear relationship
- Preserve Global Structure
- Global optimal Solution

Cons:

- High computation time
- Parameter sensitivity
- issues with complex shape: when the data lies on manifold with holes or complex structure.